The Edge of Cloud Computing

Report submitted in partial fulfillment of the requirements for the award of the Degree of

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DECLARATION

I hereby declare that **"The *Edge of Cloud Computing*"** is the result of the project work carried out by me under the guidance of ***Dr. Raju R Gondkar*** in partial fulfillment for the award of Bachelor's Degree in Commerce (Honors) by CMR University.

I also declare that this project is the outcome of my own efforts and that it has not been submitted to any other university or Institute for the award of any other degree or Diploma or Certificate.

Place: Bangalore **Name: Jonathan Samuel Robert**

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CERTIFICATE OF ORIGINALITY

(To be given by Guide and Department)

Date:

This is to certify that the Project work titled “The Edge of Cloud Computing” is an original work of Mr. Jonathan Samuel Robert; bearing University Register Number 16UG02015 and is being submitted in partial fulfillment for the award of the Bachelor’s Degree in Commerce (Honors) of CMR University. The report has not been submitted earlier either to this University / Institution for the fulfillment of the requirement of a course of study.

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DATE:

SIGNATURE OF THE SCHOOL HEAD

DATE:

ACKNOWLEDGEMENT

I would like to express my profound gratitude to our Lord Jesus Christ in bringing all the people together for this report and His guidance in bringing it to completion.

I would like to thank CMR University for providing me an opportunity to study this technology which is beyond our syllabus.

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EXECUTIVE SUMMARY

This dissertation was undertaken to understand the various benefits that edge computing brings to cloud architecture.

In cloud architecture, we face common problems such as high latency, no real-time availability of data and no real-time data transmission between IoT devices and cloud due to latency.

An edge architecture is a technology which assists cloud computing by, providing zero to low latency between devices connected to the edge and making real-time data transmission and real-time data availability possible.

As IoT (Internet of Things) devices need to connect and receive instructions from the cloud which in the real world may not be possible but with edge computing, it is possible as the edge is close to where these devices are and act as a middle man. Eliminating the need to constantly connect to the cloud, unless it is very essential to do so.

In this report, we will be looking at cloud architecture along with the edge to see how effective and efficient they work together

1. Introduction

The past decade has seen tremendous growth in the number of internet-connected devices, which has given rise to technology known as the Internet of Things (IoT). To put it simply, IoT is just a concept of inter-connecting various devices and connecting these devices to the internet. This includes everything from cell phones, coffee makers, fridge, washing machines, wearable devices, and any device you can think of that can be easily connected to any device and transfers data seamlessly.

As IoT started gaining momentum, a problem arose – that of dealing with the data from these inter-connected devices. The data we are looking at is terabytes in size. Traditionally, the data collected from these devices would be sent to the organisation's central cloud for processing. Which would take rather a long time to do so owing to the size of the data. Transferring such large datasets over the network to a central cloud can also expose sensitive organisational data to vulnerabilities on a network.

Edge computing works by storing and processing critical data on a network of micro datacentres known as the edge before it is then sent to the central cloud or datacentre repositories. This is done by bringing memory and computing power closer to the location where it is most needed. It is mainly used for dealing with IoT data, where edge devices collect the data, undertake critical processing locally and then forward the data to the cloud for storage and any further processing.

With edge computing, the IoT device will transfer the data to a local, small-form device with network, compute and storage abilities. It is this ‘micro datacentre devices that carry out the edge of network processing before moving some or all the data to the remote central repository.

The term ‘edge computing’ is essentially short for ‘edge of network computing’. It’s a process that enables the Internet of Things (IoT) devices' data to be analysed much quicker by processing it near to where it is created rather than transporting it long distances to a datacentre or a cloud. This increase in speed provides the almost real-time analysis needed by many of today’s enterprises.

Hence, we can define edge computing as a distributed IT architecture that makes it possible to process data on the border that is as close to the originating source as possible.

The name ‘edge computing’ refers to computation around the corner/edge in a network diagram. Edge computing pushes all the significant computational processing power towards the edges of the mesh. Like we said earlier as close to the originating device as possible. (picture if possible)

1.1 Edge computing explained

Edge computing works by storing and processing critical information on a network of micro datacentres before it is then sent to the central cloud or datacentre repository. Mainly used for dealing with IoT information, edge devices collect this information, undertake critical processing locally and then forward the information to the cloud for storage and any other processing furthermore.

With edge computing, the IoT device will transfer the information to a local, small form device with networking, computation and storage capabilities. It is this ‘micro datacentre’ device that carries out the edge of network processing before moving some or all the data to the remote central repository.

1.2 How does this help?

Let us consider a smart traffic light. Rather than calling home whenever is needed for data analysis. What if the device is capable of performing analytics in-house, it can accomplish a real-time analysis of streaming data and even communicate with other devices to finish tasks on the go. Edge computing, therefore, speeds up the entire analysis process, enabling quick decision-making.

Edge computing is also beneficial for the organisations as it helps them cut down costs that were earlier incurred on transferring data sets over a network. Apart from this, it also allows the organisations to filter out the useful data from the device's edge itself thereby enabling organisations to collect only valuable data and ensuring them to cut down costs on cloud computing and storage. Furthermore, edge computing also reduces the response time to milliseconds, all the while conserving the network resources. Using edge computing, we don't necessarily need to send the data over a network. Instead, the local edge computing system is responsible for compiling the data and sending frequent reports to the central cloud storage for long-term storage. By only sending the essential data to the cloud, edge computing drastically reduces the data that flows through the network.

1.3 The Importance of Edge Computing

Deploying edge devices can have numerous benefits. For example, some IoT devices can have issues with weak connectivity and this prevents them from being constantly able to send data to the central cloud. Connecting to a local device can ensure that data can continue to be processed at all times.

Another issue with connected (IoT) devices is latency over a network. When data needs to be analysed speedily, the delay in sending and receiving information can be quite problematic. By reducing the distance that this data needs to travel before processing takes place, edge computing is able to reduce latency drastically. This is essential when delays of even milliseconds are unacceptable, for example, in the monitoring of in-flight jet engines or in the operation of autonomous self-driving vehicles.

Edge computing can also reduce data traffic over a network. If an enterprise has a manufacturing plant with a vast number of sensors each producing horde amounts of data, sending all that data to the cloud would require tremendous bandwidth. Whereas, this can be dramatically reduced if most of that data could be processed at the edge of the network.

1.4 Use Cases

The deployment of Edge Computing architecture is ideal in a range of situations. One such case is when the IoT devices have weak internet connectivity, where it is not practical for them to be connected to a centralised cloud constantly.

Other such place edge computing can be used is where there's a requirement of latency-sensitive processing of data. Edge computing eliminates the factor of latency as the data does not need to be transferred over a network to central cloud storage for processing every time making it ideal for financial or manufacturing services where latencies of milliseconds are difficult to achieve.

One more use case for edge computing has been the development of the next-gen 5G cellular networks. It is predicted in this network service providers will start adding micro-data centres by either integrating into or locating adjacent to the 5G towers. Here, businesses would be able to own or rent space in these micro-data centres to perform edge computing and have direct access to a gateway into the telecom provider's central network, which can be connected to a public Infrastructure as a Service (IaaS) cloud provider.

1.4.1 Other such cases are:

- Drones - which are capable of reaching remote places that human can't even think of going. Edge computing enables these drones to review, analyse, and respond to the analysis in real-time. For instance, if a drone finds any emergency situation, it can instantaneously provide valuable information to people nearby without having first to send the data over a network and then receive the analysis.

- Augmented Reality– The introduction of edge computing has taken Augmented Reality a step further. An edge computing platform can provide highly localised data targeted at the user's point of interest, thereby enhancing the AR services.

- Automated vehicles– Giants like Google and Uber are coming up with self-driving cars. Edge computing plays a crucial role in the development of such automatic vehicles. These vehicles can process and transmit vital data in real-time to other vehicles commuting nearby using edge computing. These giants aim to make such self-driving cars a consumer reality by 2020. With the introduction of such automated vehicles, we're sure to see a decrease in the number of lives lost due to automobile accidents.

Use cases of edge computing are further discussed in greater detail in Chapter 2.

1.5 Edge Computing security

When it comes to security, there is always a mixed opinion, whether edge computing or cloud computing is more secure. Those who favour edge computing believe that its security is superior because the initial data does not need to travel over a network, thus it remains close to where it was created. At the same time, having fewer data held in a single storage location means that it is less vulnerable to man in the middle attacks. The alternative view is that data is less secure because the micro datacentre edge devices are not as securely protected as a cloud infrastructure (datacentre). As there is not as much data to lose in a single edge device, there is more chance that it will be more secure than a cloud infrastructure which is one coherent datacentre.

This leaves important security implementations for those designing and deploying edge devices and for those considering to use edge computing need to check if they are adhering to industrial standard for protection such as Virtual Private Networks (VPN), access control and data encryption.

2. Use Cases

Edge computing allows internet consumers to use more connected applications and devices without bogging down the network. It avoids network congestion by using the network's edge to bring services closer to the user than ever before. Several industries stand to benefit immensely from these potential use cases. Below is a list of some of the edge computing use cases.

2.1 5G Network Architecture

The reality is that 5G will not be able to meet the performance goals of very low latency and massive broadband without edge computing, simply because it takes time for data to travel over the fibre networks connecting the radios on the towers to the network core.

Moving the application or content closer to the radio at the edge of the network and network and latency is reduced.

Software-defined networking is one such technology that will power the move to 5G, this would require local processing to determine the best route to send data at each point of the journey.

Each node in a network can make a decision about the quality of service it needs to give a particular piece of information that comes to it, and then route that in a different way.

Prioritising important traffic can be achieved by implementing some high-performance computer hardware and radios and the 5G latency goals can be met.

In short, 5G needs edge computing.

The English city Bristol invited Android users to participate in a citywide treasure hunt that involved solving riddles. Participants experienced noticeably shorter latency times and significantly improved video streaming experiences.

2.2 AI and Virtual Assistant

Between smartphones and AI-powered virtual assistants like Amazon's Alexa and Google's Assistant, the modern household is becoming a fully integrated network on itself. As more and more of these devices enter homes, there will be a greater strain on the service provider's network as more requests flood into their servers and different forms of streaming content are delivered to users.

By incorporating edge computing architecture into their networks, companies can improve performance significantly and reduce latency. Rather than every AI virtual assistant sending processing and data requests to a centralized server, they can instead distribute the burden among edge data centres while performing some computing functions locally too.

As these edge computing use cases become more widespread, many more industries are sure to benefit from the versatility and advantages it can provide. The proliferation of localized data centres for both cloud and edge computing makes it easier than ever for organizations to expand their network reach and put themselves in a position to make the most of their data resources.

2.3 Blockchain

Distributed ledger technology like blockchain requires decentralized computing models. If you're going to do your blockchain, you need to be able to process and house these technologies locally. Each node in a blockchain is a computation unit, so blockchain isn't a centralized ledger, it's a distributed ledger. Therefore, it is also known as the edge.

2.4 Connected Vehicles

You don't want your autonomous vehicle to be asking the cloud all the time what it's supposed to be doing, or even a remote server. Hence self-driving cars need to be able to learn things without having to connect back to the cloud to process data. In order to operate safely, these vehicles will have to gather and analyse vast amounts of data pertaining to their surroundings, directions, and weather conditions, not to mention communicating with other vehicles on the road. Machine learning techniques such as reinforcement learning don't rely on training large models with big data sets which sit on the cloud instead, you can process most of this data onboard within the car. This is essentially edge computing.

With many such vehicles gathering and transmitting data, bandwidth strains are inevitable if manufacturers don’t adopt new computing solutions. It’s one thing for an office computer to experience inconvenient lag when accessing a network; it’s quite another for a self-driving car to lag when it's travelling at 100 km/h on an open highway.

The 5G Automotive Association cites edge computing as a key technology for many autonomous driving services. However, some challenges with fully autonomous vehicles can only be solved with 5G.

The connected cars can use these channels to send real-time information such as warnings about traffic safety to prevent car accidents.

Nokia participated in a 2015 test where edge computing was used over an LTE network for cars to communicate hazard information.

Edge computing architecture makes it possible for autonomous vehicles to collect, process, and share data between vehicles and to broaden networks in real time with almost no latency. Combined with a network of edge data centres geographically positioned to collect and relay critical data to municipalities, emergency response services, and auto manufacturers, edge-enabled vehicles will offer unparalleled reliability without crippling network infrastructures.

2.5 Connected homes and offices

Many people use Amazon Alexa or Google Assistant to complete tasks like turning on lights on command or changing the temperature. However, right now those tasks tend to take a few seconds to occur. With edge computing, it will be possible for them to happen in near real-time.

Urban areas are quickly becoming massive information gathering centres, with sensors collecting data on traffic patterns, utility usage, and key infrastructure every single day. While that data allows city officials to respond to problems faster than ever before, all of that information must be collected, stored, and analysed before it can be put to use. Traditional cloud solutions aren't able to provide immediate response times for the multitude of devices operating on the outskirts of the network.

Edge computing architecture makes it possible for devices regulating utilities and other public services to respond to changing conditions in near real-time. This combined with the rising number of autonomous vehicles and the ever-expanding internet of things, smart cities have the potential to transform how people live and utilize services in an urban environment. Since all edge computing use cases rely upon devices collecting data to carry out basic processing tasks, the city of the future will have the ability to react dynamically to changing conditions as they occur.

2.6 Deployment of Enterprise and Campus Networks

A reliable, readily available network is crucial for large enterprises and campus networks. Too many people using the same network can result in high latency for users. Edge computing resolves that. In an enterprise network, edge computing allows a ton of employees to simultaneously access the company's intranet. Companies can complete mass training without halting network speed.

2.7 Fog Computing

Fog computing is an architecture that uses edge devices to connect to a distributed computing model routed over the internet backbone. Distributed computing systems carry out a substantial amount of computation, storage, communication across the edge and it is a continuum to the cloud.

2.8 Financial Sector

Banking institutions are adopting edge computing combined with smartphone apps to better target services to customers. They're also incorporating the same principles to provide ATMs and other such kiosks with the ability to gather and process data locally, making them more responsive and allowing them to offer a broader suite of features.

Finance firms dealing in hedge funds and other such high-volume markets, where even a millisecond of lag in a trading algorithm computation can mean a substantial loss of money. Edge computing architecture allows them to place servers in data centres near such stock exchanges around the world to run resource-intensive algorithms as close to the source of data as possible. This provides them with the most accurate and up to date information to keep their business moving.

2.9 Healthcare

The healthcare industry has long struggled to integrate the latest IT solutions, but edge computing offers exciting new possibilities for delivering patient care. With IoT devices capable of delivering vast amounts of patient-generated health data (PGHD), healthcare providers could potentially have access to critical information about their patients in real time rather than interfacing with slow and incomplete databases. Medical devices themselves could also be made to gather and process data throughout the course of diagnosis or treatment.

Edge computing could make a significant impact on the delivery of healthcare services to hard-to-reach rural areas. Patients in these regions are often many miles from the nearest health provider and even if a healthcare professional evaluates them on-site, they may not be able to access crucial medical records. With edge computing, devices could gather, store, and deliver that information in real time, and even use their processing capabilities to recommend treatments.

While regulatory requirements for the sharing and disclosure of medical information would make any edge solution challenging to implement, other emerging security measures such as blockchain technology could provide new ways to address such concerns.

2.9.1 Telemedicine

Glucose monitors, at-home personal health-monitoring devices, and fitness bands like Fitbit are reshaping the medical field. Data from these devices can be used to instantly update a person’s medical records and provide caregivers, physicians, and emergency personnel with accurate, up-to-date patient information.

2.10 Industrial automation

Edge computing can help create machines that can sense, detect, and learn things without having to be programmed by using machine learning.

For example, if the sun shining through a window hits a machine for part of the day, the machine will eventually be able to tell that the temperature change doesn't mean that something is wrong.

In industrial processes, machines generally need to be adjusted based on the environment or the quality of materials coming in. If one is monitoring the process closely and locally, you can extend the life of that machine and extend the operational efficiency of that machine by tweaking what it's doing, edge computing can help detect such dangers and can help find the right fix for them. Alerts for what's happening with a machine are best done when processed close to that machine.

2.10.1 Industrial Manufacturing

Perhaps no industry stands to benefit more from IoT devices than the manufacturing sector. By incorporating data storage and computing into industrial equipment, manufacturers can gather data that will allow for better predictive maintenance and energy efficiency, allowing them to reduce costs and energy consumption while maintaining better reliability and productive uptime. Smart manufacturing techniques informed by ongoing data collection and analysis will also help companies to customize production runs to better meet consumer demands.

Edge computing can also provide great advantages to industries operating where bandwidth is low or non-existent. Offshore oil rigs, for instance, can utilize edge computing architecture to gather, monitor, and process data on a variety of environmental factors without having to depend upon a distant data centre in the cloud.

2.10.2 Remote monitoring in industry

Remote monitoring will be invaluable for enterprises that operate in remote regions. With edge computing, these businesses will be able to respond to new events, such as a busted pipe on an oil rig, in real time. Immediate on-the-ground responsiveness would be delayed if the site data needed to travel to a traditional data centre hundreds of miles away to report such issues.

Likewise, trucking companies will be better able to manage their fleets to effectively service their trucks, maximizing uptime and lowering costs.

And restaurants and food service companies can receive text-based alerts when sensors detect a change in refrigeration temperatures, ensuring refrigerators are at an optimal temperature at all times.

2.11 Retail

Several retail chains such as Amazon Go, are creating more immersive in-store environments with technologies like AR to attract additional shoppers. This requires lower latency, which is where edge computing capabilities come in.

2.12 Video Optimisation

Video cameras can gather gigabytes of data at a time. Transporting all of that data to a remote processor takes a lot of time and money particularly while using motion detection or facial tracking features. But edge computing can handle the sort of detections that would traditionally have to be done on a large-scale computer.

Video buffering is caused by the Transmission Control Protocol (TCP) not adapting fast enough to varying radio conditions in an LTE network can be surpassed by using 5G networks utilising the edge computing technology. This technology dodges those video streaming issues by communicating to the video server the best bit rate for the given radio conditions. This reduces the buffering time of the device's video stream.

2.13 Virtual Reality & Augmented Reality (VR/AR)

Both of these applications require low latency and real-time response times to function. This use case is commonly referred to in discussions about 5G as well.

VR renders an immersive environment of digital graphics, while AR transposes digital graphics onto a real environment. Companies have already incorporated such technologies into their marketing such as Ikea, which created a virtual reality app.

AR and VR tools in conjunction with each other can also be used for employee training to help them understand the environment around.

While virtual reality may be a more familiar term to most people, augmented reality (AR) is both more common and has more practical applications. Instead of generating an entirely virtual world, AR overlays digital elements over real-world environments. Wearable AR devices like glasses and headsets are sometimes used to create this effect, but most users have experienced AR through their smartphone displays. Anyone who has played games like Pokémon GO or used a filter on Snapchat or Instagram has made use of AR.

The technology behind AR requires devices to process visual data and incorporate pre-rendered visual elements in real time. Without edge computing architecture, this visual data would need to be delivered back to centralized cloud servers where the digital elements could be added before sending it back to the device. This arrangement would inevitably result in significant latency. Edge computing allows IoT devices to composite AR displays instantly, allowing users to look anywhere to take in new AR details without having to deal with loading times.

Conclusion:

Enterprises have focused on cloud computing during the last decade, but increasingly they are shifting gears and giving serious consideration to edge computing, which provides benefits like low latency and real-time processing.

3. Literature Review

3.1.1 Edge computing technologies for Internet of Things: a primer (2018)

With the rapid development of mobile internet and Internet of Things applications, the conventional centralized cloud computing is encountering severe challenges, such as high latency, low Spectral Efficiency (SE), and nonadaptive machine type of communication. Motivated to solve these challenges, new technology is driving a trend that shifts the function of centralized cloud computing to edge devices of networks. Several edge computing technologies originating from different backgrounds to decrease latency, improve SE, and support the massive machine type of communication have been emerging. This paper comprehensively presents a tutorial on three typical edge computing technologies, namely mobile edge computing, cloudlets, and fog computing. In particular, the standardization efforts, principles, architectures, and applications of these three technologies are summarized and compared. From the viewpoint of a radio access network, the differences between mobile edge computing and fog computing are highlighted, and the characteristics of fog computing-based radio access network are discussed. Finally, open issues and future research directions are identified as well.

3.1.2 Edge Computing: Vision and Challenges (2017)

Mahadev Satyanarayanan, School of Computer Science, Carnegie Mellon University

Edge computing is a new paradigm in which the resources of a small data centre are placed at the edge of the Internet, in close proximity to mobile devices, sensors, and end users. Terms such as "cloudlets," "micro data centres," "fog, nodes" and "mobile edge cloud" have been used in the literature to refer to these edge-located computing entities. Located just one wireless hop away from associated mobile devices and sensors, they offer ideal placement for low-latency offload infrastructure to support emerging applications. They are optimal sites for aggregating, analysing and distilling bandwidth-hungry sensor data from devices such as video cameras. In the Internet of Things, they offer a natural vantage point for organizational access control, privacy, administrative autonomy and responsive analytics. In vehicular systems, they mark the junction between the well-connected inner world of a moving vehicle and its tenuous reach into the cloud. For cloud computing, they enable fallback cloud services in hostile environments. Significant industry investments are already starting to be made in edge computing. This talk will examine why edge computing is a fundamentally disruptive technology and will explore some of the challenges and opportunities that it presents to us.

3.1.3 Edge Computing: Needs, Concerns and Challenges (2017)

In numerous parts of computing, there has been a continuous issue between the centralization and decentralization aspect which prompted to move from mainframes to PCs and local networks in the past, and union of services and applications in clouds and data centres. The expansion of technological advances such as high capacity mobile end-user devices, powerful dedicated connection boxes deployed in most homes, powerful wireless networks, and IoT (Internet of Things) devices along with developing client worries about protection, trust and independence call for handling the information at the edge of the network. This requires taking the control of computing applications, information and services away from the core to the other the edge of the Internet. The relevance of cloud computing to mobile networks is on an upward spiral. Edge computing can possibly address the concerns of response time requirement, bandwidth cost saving, elastic scalability, battery life constraint, QoS, etc. MEC additionally offers high bandwidth environment, ultra-low latency that gives real-time access to radio networks at the edge of the mobile network. Currently, it is being used for enabling on-demand elastic access to, or an interaction with a shared pool of reconfigurable computing resources such as servers, peer devices, storage, applications, and at the edge of the wireless network in close proximity to mobile users. It overcomes obstacles of traditional central clouds by offering wireless network information and local context awareness as well as low latency and bandwidth conservation. In this paper, we introduce edge computing and edge cloud, followed by why do we need edge computing, its classifications, various frameworks, applications and several case studies. Finally, we will present several challenges, concerns and future scope in the field of edge computing.

3.1.4 Challenges and Opportunities in Edge Computing (2016)

Many cloud-based applications employ a data centre as a central server to process data that is generated by edge devices, such as smartphones, tablets and wearables. This model places ever-increasing demands on communication and computational infrastructure with an inevitable adverse effect on Quality of Service and Experience. The concept of Edge Computing is predicated on moving some of this computational load towards the edge of the network to harness computational capabilities that are currently untapped in edge nodes, such as base stations, routers and switches. This position paper considers the challenges and opportunities that arise out of this new direction in the computing landscape.

3.1.5 Edge Computing: Vision and Challenges (2016)

The proliferation of the Internet of Things (IoT) and the success of rich cloud services have pushed the horizon of a new computing paradigm, edge computing, which calls for processing the data at the edge of the network. Edge computing has the potential to address the concerns of response time requirement, battery life constraint, bandwidth cost saving, as well as data safety and privacy. In this paper, we introduce the definition of edge computing, followed by several case studies, ranging from cloud offloading to smart home and city, as well as collaborative edge to materialize the concept of edge computing.

Finally, we present several challenges and opportunities in the field of edge computing and hope this paper will gain attention from the community and inspire more research in this direction.

3.1.6 Research on Edge Computing: A Detailed Study (2016)

The success of rich cloud services has pushed the horizon of a new computing paradigm, Edge computing, which calls for processing the data at the edge of the network. Edge computing has the potential to address the concerns of response time requirement, battery life constraint, bandwidth cost saving, as well as data safety and privacy. In this paper, we introduce the definition of Edge computing, followed by several case studies as well as collaborative Edge to materialize the concept of Edge computing. Finally, we present several challenges and opportunities in the field of Edge computing.

3.1.7 The Emergence of Edge Computing (2016)

Industry investment and research interest in edge computing, in which computing and storage nodes are placed at the Internet's edge in close proximity to mobile devices or sensors, have grown dramatically in recent years. This emerging technology promises to deliver highly responsive cloud services for mobile computing, scalability and privacy-policy enforcement for the Internet of Things, and the ability to mask transient cloud outages.

3.1.8 Edge-centric Computing: Vision and Challenges (2015)

In many aspects of human activity, there has been a continuous struggle between the forces of centralization and decentralization. Computing exhibits the same phenomenon; we have gone from mainframes to PCs and local networks in the past, and over the last decade we have seen a centralization and consolidation of services and applications in data centres and clouds. We position that a new shift is necessary. Technological advances such as powerful dedicated connection boxes deployed in most homes, high capacity mobile end-user devices and powerful wireless networks, along with growing user concerns about trust, privacy, and autonomy requires taking the control of computing applications, data, and services away from some central nodes (the “core") to the other logical extreme (the “edge") of the Internet. We also position that this development can help to blur the boundary between man and machine, and embrace social computing in which humans are part of the computation and decision-making loop, resulting in a human cantered system design. We refer to this vision of human cantered edge-device based computing as Edge-centric Computing. We elaborate in this position paper on this vision and present the research challenges associated with its implementation.

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3.2 Need for the study

The Internet of Things (IoT) is way more than just connecting devices to the cloud through the internet, but it is about generating new business insights, automating production processes and ultimately accelerating innovation cycles.

With the number of use cases of such IoT connected devices, which show no sign of slowing down in the near future. As per statistics, these devices are expected to reach 8 billion by 2020, which in turn would generate massive amounts of data. The capability to store and move such data is, however, becoming difficult.

In order to efficiently capture, transmit and process, analysis and store such high volumes of data have to happen at the edge of the wide area network (WAN), simply put the edge of the internet.

Hence Edge Computing helps address such issues. They are:

- To reduce the amount of data transmitted and stored in the cloud
- To reduce the lag time in data transmission and processing
- To reduce the signal to noise ratio

3.3 Objectives of the study

The various objectives are as follows:

- To study Low Latency between the consumer and the edge computing.
- To study Real Time Availability of information between two devices connected within an edge deployment.
- To Study Real-Time Data Transmission between the edge and the cloud.
- To study how to bring company and customer together.
- To study how to provide good digital experience to consumers.
- To study how to increase productivity using an edge infrastructure.

3.4 Limitations of the study

Some of the limitations of this are mentioned below:

- There is no robust redundancy like the cloud, within an edge infrastructure.
- The potential loss of data or corruption of data due to limited to no redundancy.
- Longer outage time as the edge nodes are not placed with a particular datacentre, but at the edge of a network.
- Lack of tier III and tier IV security which is an essential part of a datacentre.

3.5 Data collection

Data collection for this project report is obtained from secondary sources such as websites, blogs and statistical websites and other such institutions who publish secondary data.

3.6 Operational concepts

3.6.1 Tier III Datacentre

In India, a tier III datacentre comprises of 99.982 per cent of availability, with interruption changes at 1.6 hours a year and having “N+1” need for power supply. Where “N” denoted the amount of power supply required and “+1” denotes the backup option readily available during a failover.

3.6.2 Tier IV Datacentre

In India, a tier IV datacentre comprises of 99.995 per cent of availability, with interruption changes at 0.8 hours a year and having “N+1” need for power supply. Where “N” denotes the amount of power supply required and “+1” denotes the backup option readily available during a failover.

4. Data Analysis and Interpretation

4.1 Introduction

This chapter talks about data analysis and interpretation of the secondary data collected from sites like Sam solutions, Statista, Wikibon & Others. Who have helped in visualising data pertaining to the growth of IoT devices, advantages of edge computation and how cost efficient it is to combine edge computing with the present and existing cloud infrastructure.

4.2 Number of connected devices worldwide

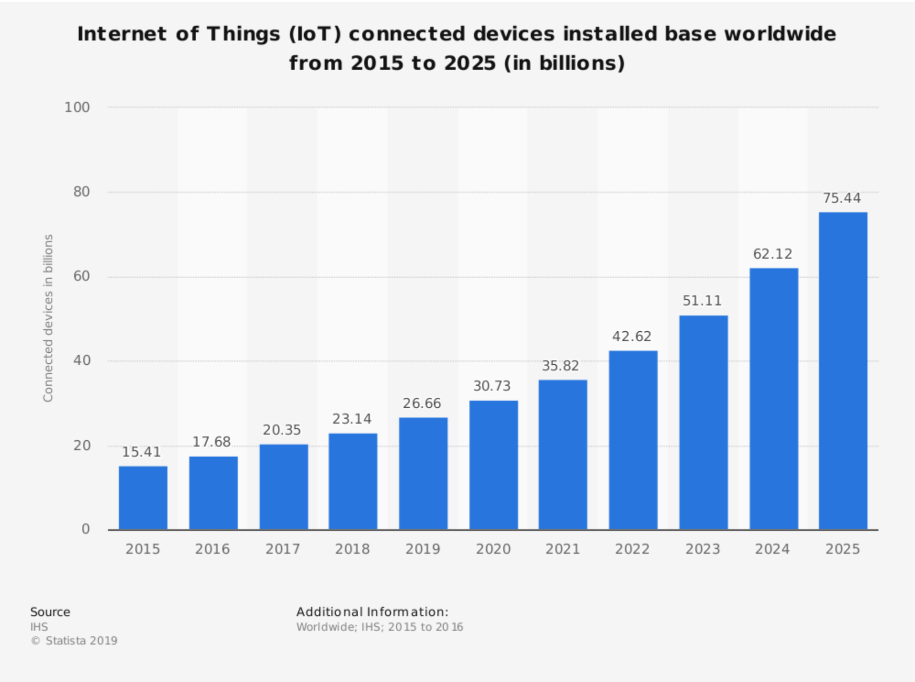


Figure 1: IoT connected devices installed base worldwide from 2015 to 2025

As per the figure above, it is estimated that by 2020, there will be over 30 billion IoT devices worldwide, and in 2025, the number will exceed 75 billion connected things, according to Statista. All these devices will produce huge amounts of data that will have to be processed quickly and in a sustainable way.

To meet the growing demand for IoT solutions, technologies such as edge and fog computing comes into action which is on par with cloud computing. Where Edge and Fog are better at some things.

4.3 Comparison of Cloud and Edge Computing

Requirements	Cloud Computing	Edge Computing
Geo-distribution	Centralised	Distributed
Distance between client and server	Multiple hops	One hop
Latency	High	Low
Delay Jitter	High	Very low
Location Awareness	No	Yes
Mobility Support	Limited	Supported
Location of service	Within the Internet	At the edge

Table 4.1: Comparison of Cloud Computing and Edge Computing

4.4 Windfarm Case Study

To conduct this study two services are chosen they are:

a. AWS's IoT services
b. Pivot3 SAN (Storage Attached Network) Server with Time-series Database

These services will be using both cloud as well as cloud plus edge technologies to compare various differences among them.

This analysis is on a small remote wind farm with the consolidation of security sensors apart from security cameras, along with sensors on the wind turbine and access sensors for all the employee's entry and exit at a physical terminal. The number of sensors and video surveillance cameras is expected to grow over time, much larger than this pilot system. By no means is this model of a complete long-term sensor coverage of the wind farm.

The details of the initial case study are given below in Table 4.2 Cost comparison of Cloud only vs Edge & Cloud Management

Cost Comparison of Cloud-only Management & Processing vs. Edge + Cloud							
		Cloud-only Processing			Edge + Cloud Processing		
		# Sensors	Video (MPEG)	Total	# Sensors	Video (MPEG)	Total
	# Inputs	100	2		100	2	
Edge Transport Costs	Readings/minute	20	60		20	60	
	Bytes/Read	20	672		20	672	
	GB/Month	1.7	3.5	5.2	1.7	3.5	5.2
	3-year GB	62	125	188	62	125	188
	Local Consolidation	0%	0%		95%	95%	
	Miles	200	200		200	200	
	$/GB/Month/Mile	$2	$2		$2	$2	
	Monthly Cost	$691	$1,393		$35	$70	
	Edge 3-year Transport Costs	$24,883	$50,165	$75,048	$1,244	$2,508	$3,752
Cloud Costs	AWS $/Million Messages	$5	$5		$5	$5	
	AWS Message Size (Bytes)	512	512		512	512	
	AWS Cloud Cost/Month	$86	$10		$4	$1	
	AWS 3-year Cloud Costs	$3,110	$373	$3,484	$156	$19	$174
	On-site Equipment 3-year Costs			$2,000			$25,000
	Total 3-year Costs			$80,531			$28,927
Source: © Wikibon IoT Project. Reference Models for Cloud is AWS IoT Service. Reference Model for On-site Server SAN is Pivot3 with an Open Source Time Series Database On-site & in the AWS Cloud							

Table 4.2 Cost Comparison of Cloud only Management vs. Edge plus Cloud Management

As mentioned above the distance from the wind farm to the datacentre is assumed to be 200 miles (approximately 322 kilometres). The video from the surveillance is assumed to be compressed by about 100:1 ratio in MPEG [*Moving Picture Experts Group*] video format. This is the minimum-security quality standard which would not provide enough data from the footage to run advance functions such as face recognition.

The sensors data shows a small amount of data (i.e. 20 bytes) gathered every three seconds from 100 sensors. The amount of sensor data is about half the size of the two video feeds coming from the surveillance which is low quality (i.e. 100:1 compression ratio) in MPEG format or codec.

The key differences between the red and the blue sections in Table 4.2 are the percentage of local consolidation.

The red section assumes there is no local consolidation (or time-series database) and all the data is processed and managed in the cloud itself.

The blue section assumes edge computation with 20:1 reduction in data required to be transmitted (to the cloud). This is easily achieved in reality and higher data reductions of 100:1 or more are easily achievable.

In Figure 2 shown below takes the data in Table 4.2 and shows the cost results for Edge computing reduction percentages from zero all the way to ninety-nine. The cross over point for edge is at 30% reduction as per this figure.

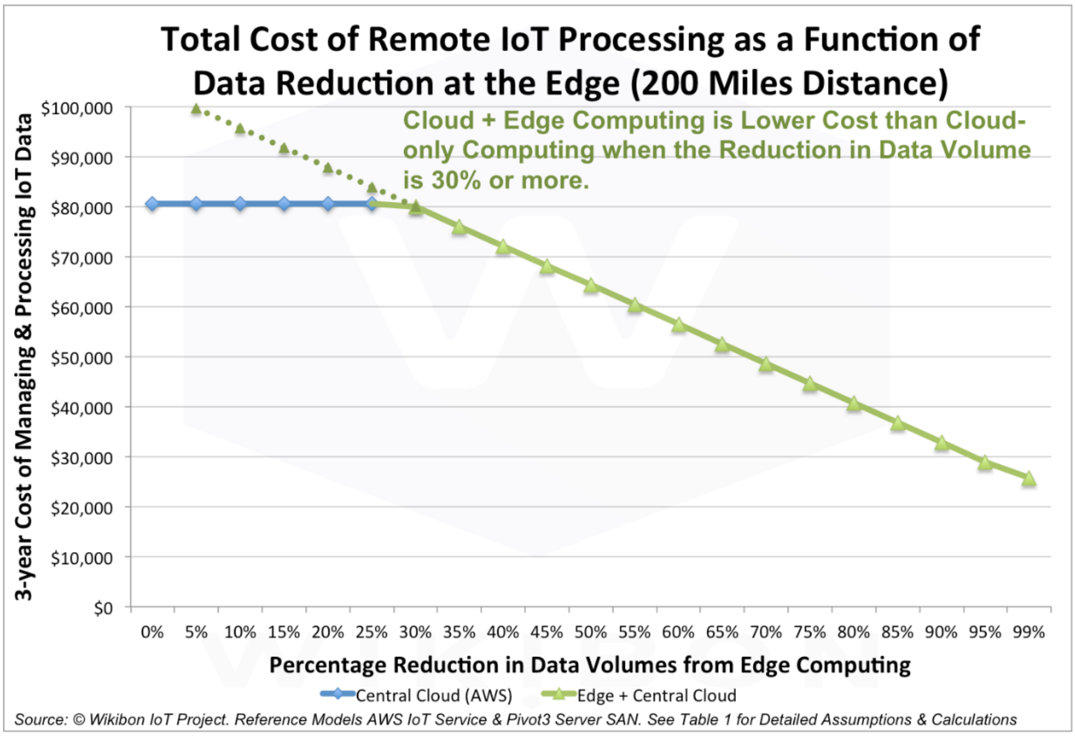


Figure 2: Total Cost of Remote IoT Processing [Distance: 200 Miles]

This representation is only an initial deployment with minimalistic data generated.

Edge computing at higher reduction levels of 100:1 or higher would allow much richer data to be gathered and processed at the Edge which in turn would provide a safer, more reliable and lower cost solution.

Figure 3 shows the calculations for a 100-mile distance between the Edge and the datacentre. Even with shorter distances, the advantages of Edge computing are astonishing.

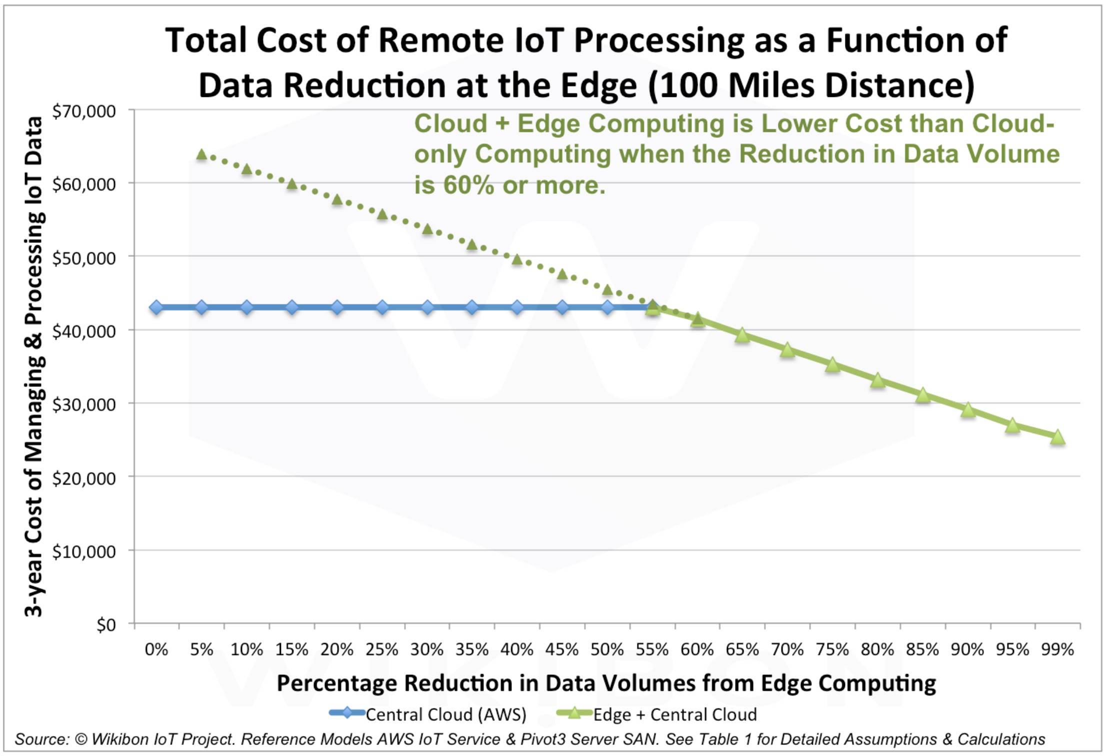


Figure 3: Total Cost of Remote IoT Processing [Distance: 100 Miles]

The above figure shows a breakeven at 60% data reduction when the distance between the Edge and a datacentre is of 100 miles (approximately 161 kilometres).

4.5 Conclusion

As per the analysis carried on the data mentioned above, we can conclude by saying IoT systems will be safer, more reliable, have lower cost and be more functional using an Edge computing plus Cloud computing approach.

The summary of findings of the analysis in this chapter is found in Chapter 5 along with an estimation of how much data is generated by category, a graph between 2011-2020.

5. Summary of Findings, Suggestions and Conclusion

5.1 Summary of findings

With reference to the previous chapter, the analysis on the remote windfarm with security cameras and other sensors was undertaken to compare cloud-only architecture approach with a low-cost converged working in unison with cloud computing (i.e. edge computing)

Figure 4 shown below is a summary of the findings of the data analysis undertaken in the previous chapter. It compares the 3-year management costs and well as processing costs of a cloud only solution using AWS IoT services compared with an Edge plus Cloud solution using a Pivot3 Server SAN with an Open Source Time-series Database, referred to as edge. Together with AWS IoT services.

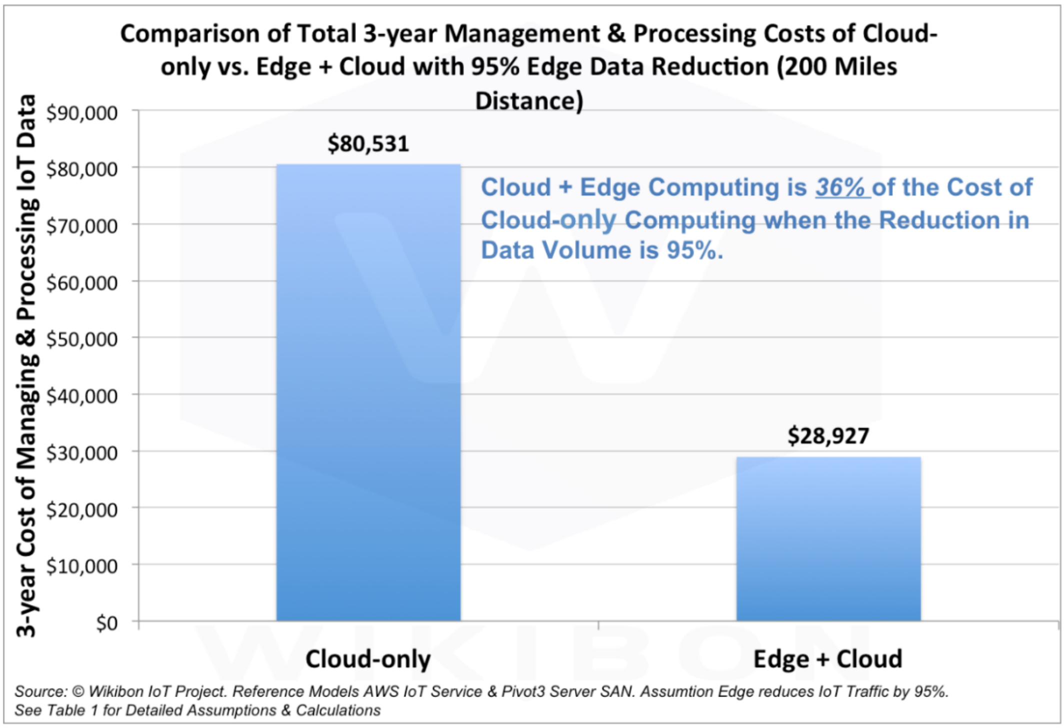


Figure 4: Comparison of Total 3-year Management & Processing Costs of Cloud Computing and Cloud plus Edge Computing with 95 per cent in the reduction of IoT traffic.

With a distance of 200 miles (approximately 322 kilometres) between the wind farm and the cloud. Assuming a 95 per cent reduction in traffic of IoT devices from using the edge computing techniques.

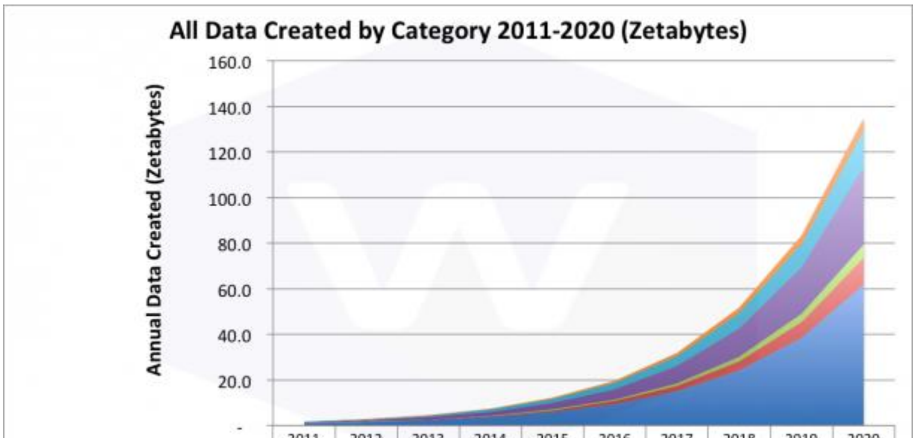
Total cost is reduced from $81,000 to $29,000 over 3 years. The cost of Edge plus Cloud computing with a 20:1 reduction in data transmitted is around 1/3th the cost of a cloud-only deployment.

Table 4.2 provides detailed assumptions and calculations for this summary of the findings mentioned above.

5.1.1 Server SAN

Server SAN is a rapidly rising architecture or software led enterprise systems. Moving what was done is storage arrays. Adoption of Server SAN will lead to lower cost and improved performance. For IoT Edge processing, Server SAN is lowering the cost and is much more efficient than other traditional systems.

Pivot3 is a major contributor to the migration of video surveillance from the analogue to the digital spectrum. As per Figure 5 shown below, surveillance is to be about 46 per cent of the data created by 2020. That is around 62 Zettabytes, *where 1 Zettabyte is equal to 1 billion Terabytes or a trillion Gigabytes.* As the IoT architecture develops, most of the IoT Edge sites will combine sensor data with surveillance data, this would ultimately add to the storage of video surveillance footage. Hence having an architecture like Server SAN in the Edge is likely to be a tremendous advantage in the long run.



	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Consumer Images, Voice & Video	0.1	0.2	0.3	0.4	0.6	1.0	1.5	2.3	3.5	5.4
Entertainment & Social Media	0.3	0.5	0.7	1.1	1.7	2.7	4.3	6.7	10.4	16.2
Data Processing	0.4	0.6	1.0	1.7	2.8	4.6	7.6	12.5	20.5	33.7
Medical	0.0	0.1	0.1	0.2	0.4	0.7	1.2	2.0	3.5	6.1
Internet of Things	0.1	0.2	0.3	0.5	0.8	1.4	2.3	4.0	6.8	11.5
Surveilance	0.9	1.4	2.3	3.7	5.9	9.5	15.1	24.2	38.8	62.0

Figure 5: All Data Created by Category

5.2 Suggestions

AWS is philosophically opposing towards Edge computing but were the among the first to support Edge models with AWS IoT Greengrass along with Microsoft Azure's IoT edge. Other cloud services like Google Cloud, Alibaba, General Electric and many others are likely to provide superior Edge computing with Cloud solutions in the future to compete in the same domain.

Internet of Things will develop with an Edge + cloud computing architecture for the vast majority of sensor implementations. The local Edge computing will almost always include a time-series database to assist in data reduction and local and cloud processing. The majority of the IoT processing will be machine to machine (M2M), without the slow end-user response times present in most client/server systems.

Machine to machine (M2M) interaction will demand very low response times from all the compute system elements, which will not be possible over long distances is what Wikibon believes as per there analysis.

As per the analysis in the previous chapter, we believe that Edge computing will be a software-led model. There are two distinct types of solution in this space:

- Edge systems based on Intel processors and a new sensor ecosystem.
- Edge systems based on ARM processors, utilizing the existing mobile sensor ecosystem.

Major potential suppliers are Android systems from Google and Samsung, and iOS systems from Apple.

It is unclear which model will dominate. There is certainly room for each solution type to be dominant in different segments of the market, let us assume ARM being dominant in consumer “set-top box” type home systems and Intel being dominant in industrial sites. This would lead to the reduction in the cost of Edge compute systems, with set-box type home systems coming in at well below $1,000 in the near future, as mentioned by Wikibon

Wikibon believes that projections based on all sensor traffic going over the internet are unlikely to happen in the long run. The impact of Edge IoT computing is very likely to significantly reduce the amount of internet traffic required.

There is enormous potential for higher level analytics and automation to be processed in the cloud, with the ability to analyse and optimize IoT inputs from many different sources using edge architecture.

5.3 Conclusion

Edge computing is an exciting development in network infrastructure that is only beginning to realise its potential. Enterprises have focused on cloud computing during the last decade, but increasingly they are shifting gears and giving serious consideration to edge computing, which provides benefits like low latency and real-time processing.

In IoT environments, when vast amounts of data are collected in real time, moving such massive amounts of data to a processing location can be a tremendous challenge. A key capability for edge computing is micro-modular data centres. These small, typically cabinet-size data centres bring computing and storage capacity close to the source of the data, so it can be quickly and securely processed, analysed and stored.

The greatest benefit of this architecture is its speed and quickness, which can also aid data analysis without burdening the Cloud networks.

Even with benefits like increased speed, optimisation and outage reduction, adoption of edge computing will need some awareness. Look at how long cloud adoption took. But over time, businesses will learn how edge computing can speed up operations while reducing the usual risk factors.

With its implementation business can think of offering much more services than ever before such as hyper personalised shopping experiences to their customers.

Finally, an enterprise's performance on the edge depends, on the quality of its hardware. For robust networking, enterprises are increasingly switching to Software Defined Wide Area Networks (SD-WANs) due to their ability to dramatically reduce network interruptions, ensuring high performance consistently for business-critical applications, and improve network efficiency and expandability.

Bibliography

6 use cases for edge computing. (2018, November 30). Retrieved 14 April 2019, from Sprint Business website: https://business.sprint.com/blog/use-cases-for-edge-computing/

A Beginners Guide to Edge Computing | upGrad blog. (n.d.). Retrieved 9 February 2019, from https://www.upgrad.com/blog/a-beginners-guide-to-edge-computing/

An Introduction to Edge Computing for Enterprises | Web Hosting UK Blog. (n.d.). Retrieved 13 March 2019, from https://www.webhosting.uk.com/blog/an-introduction-to-edge-computing-for-enterprises/

Baker, J. (2017, November 5). Edge Computing — The New Frontier of the Web. Retrieved 13 March 2019, from Hacker Noon website: https://hackernoon.com/edge-computing-a-beginners-guide-8976b6886481

Be smart about edge computing and cloud computing | InfoWorld. (n.d.). Retrieved 9 February 2019, from https://www.infoworld.com/article/3288604/cloud-computing/be-smart-about-edge-computing-and-cloud-computing.html

Bhattacharyya, S. (2016). Research on Edge Computing: A Detailed Study. International Journal of Information Technology, 2(6), 5.

Demystifying Edge vs. Cloud Computing - DZone IoT. (n.d.). Retrieved 14 April 2019, from https://dzone.com/articles/demystifying-the-edge-vs-cloud-computing

DeNisco-Rayome, A. (n.d.). Ten scenarios where edge computing can bring new value. Retrieved 14 April 2019, from ZDNet website: https://www.zdnet.com/article/10-scenarios-where-edge-computing-can-bring-new-value/

Edge computing. (2019). In Wikipedia. Retrieved from https://en.wikipedia.org/wiki/Edge_computing

Edge computing technologies for Internet of Things: a primer - ScienceDirect. (n.d.). Retrieved 23 April 2019, from https://www.sciencedirect.com/science/article/pii/S2352864817301335

Felter, B. (n.d.). 5 Edge Computing Use Cases with Huge Potential. Retrieved 14 April 2019, from https://www.vxchnge.com/blog/edge-computing-use-cases

Five Edge Computing Use Cases. (n.d.). Retrieved 14 April 2019, from SDxCentral website: https://www.sdxcentral.com/edge/definitions/multi-access-edge-computing-use-cases/

Fog Computing vs. Cloud Computing. What are the Key Differences? | SaM Solutions. (n.d.). Retrieved 14 April 2019, from https://www.sam-solutions.com/blog/fog-computing-vs-cloud-computing-for-iot-projects/

freeCodeCamp.org. (n.d.). Beyond the Cloud: Edge Computing. Retrieved from https://www.youtube.com/watch?v=SQipnBNVjv0&feature=youtu.be

Garcia Lopez, P., Montresor, A., Epema, D., Datta, A., Higashino, T., Iamnitchi, A., … Riviere, E. (2015). Edge-centric Computing: Vision and Challenges. ACM SIGCOMM Computer Communication Review, 45(5), 37–42. https://doi.org/10.1145/2831347.2831354

Ghosh, P. (2018, July 26). Internet of Things vs. Edge Computing: Processing Real-Time Data. Retrieved 23 April 2019, from DATAVERSITY website: https://www.dataversity.net/internet-things-vs-edge-computing-processing-real-time-data/

Hosac, K. (2017, December 22). Three Reasons Why You Need Edge Computing Now. Retrieved 23 April 2019, from DATAVERSITY website: https://www.dataversity.net/three-reasons-need-edge-computing-now/

IIoT Edge Computing vs. Cloud Computing. (2017, April 26). Retrieved 23 April 2019, from Open Automation Software website: https://openautomationsoftware.com/blog/iiot-edge-computing-vs-cloud-computing/

Introduction to Edge Computing in IIoT. (n.d.). 19.

IoT: number of connected devices worldwide 2012-2025 | Statista. (n.d.). Retrieved 14 April 2019, from https://www.statista.com/statistics/471264/iot-number-of-connected-devices-worldwide/

Journal, I. (n.d.). Fog Computing: Synergizing Cloud, Big Data and IoT-Strengths, Weaknesses, Opportunities and Threats (SWOT) Analysis. Retrieved from https://www.academia.edu/34458128/Fog_Computing_Synergizing_Cloud_Big_Data_and_IoT-Strengths_Weaknesses_Opportunities_and_Threats_SWOT_Analysis

NTT SCL. (n.d.). Edge Computing Platform. Retrieved from https://www.youtube.com/watch?v=RjMS15V_7nQ&feature=youtu.be

RESEARCH PAPER: Edge Computing: The Fourth Wave Rises | Moor Insights & Strategy. (n.d.). Retrieved 23 April 2019, from http://www.moorinsightsstrategy.com/research-paper-edge-computing-the-fourth-wave-rises/

Satyanarayanan, M. (2017). Edge Computing: Vision and Challenges. Retrieved from https://www.usenix.org/conference/hotcloud17/program/presentation

Sharma et al. - 2017 - Edge Computing Needs, Concerns and Challenges.pdf. (n.d.). Retrieved from https://www.ijser.org/researchpaper/Edge-Computing-Needs-Concerns-and-Challenges.pdf

Sharma, V., Rai, V. P., & Sharma, D. K. K. (2017). Edge Computing: Needs, Concerns and Challenges. 8(4), 13.

Shi, W., Cao, J., Zhang, Q., Li, Y., & Xu, L. (2016). Edge Computing: Vision and Challenges. IEEE Internet of Things Journal, 3, 1–1. https://doi.org/10.1109/JIOT.2016.2579198

The Impact of Augmented Reality on Retail | Retail Perceptions. (n.d.). Retrieved 14 April 2019, from http://www.retailperceptions.com/2016/10/the-impact-of-augmented-reality-on-retail/

The role of Edge Computing in 5G. (2018, November 27). Retrieved 14 April 2019, from Data Economy website: https://data-economy.com/the-role-of-edge-computing-in-5g/

The Vital Role of Edge Computing in the Internet of Things - Wikibon Research. (n.d.). Retrieved 14 April 2019, from https://wikibon.com/the-vital-role-of-edge-computing-in-the-internet-of-things/

Varghese, B., Wang, N., Barbhuiya, S., Kilpatrick, P., & Nikolopoulos, D. S. (n.d.). Challenges and Opportunities in Edge Computing. 6.

Varghese et al. - Challenges and Opportunities in Edge Computing.pdf. (n.d.). Retrieved from https://arxiv.org/pdf/1609.01967.pdf

What Edge Computing Means for Infrastructure and Operations Leaders. (n.d.). Retrieved 14 April 2019, from http://gtnr.it/2gopHMQ

Why Do We Need Edge Computing? | Cloud Computing. (2018, August 16). Retrieved 23 April 2019, from IoT For All website: https://www.iotforall.com/edge-computing-benefits/

Will Edge Computing devour the Cloud? (2017, September 14). Retrieved 23 April 2019, from We speak IoT website: https://www.wespeakiot.com/will-edge-computing-devour-cloud/

WMedia. (2018, May 7). Living on The Edge. Retrieved 23 April 2019, from W Media website: https://w.media/living-on-the-edge/